Malliavin Calculus for Greeks Computation in Finance

Monte Carlo Estimation for European, Asian, and Exotic Derivatives

Extensions to the Down-and-Out Barrier Call and a Finite-Difference Sensitivity Analysis

Mauricio Silva López

Actuarial Science & Mathematics, UNAM – FES Acatlán / Facultad de Ciencias
Data Operations Intern, WTW Mexico

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Abstract

We implement and extend the Malliavin Calculus framework for computing option sensitivities (Greeks) via Monte Carlo simulation, following the foundational exposition of Montero & Kohatsu-Higa [1]. Three original contributions supplement the reference paper: *(i)* a full convergence and variance analysis for Gamma—the second-order Greek that the original article omits despite its practical importance; *(ii)* a finite-difference sensitivity study quantifying the bias–variance trade-off as the bump size ε varies, which motivates the Malliavin approach from a statistical-efficiency perspective; and *(iii)* the extension of the Malliavin integration-by-parts formula to a Down-and-Out Barrier Call, a path-dependent exotic derivative not covered

in [1]. All results are reproduced numerically in a companion Jupyter notebook under the Black-Scholes model.

1 Introduction

Pricing and risk management of derivative securities requires computing the sensitivities of an option price P with respect to its model parameters. These quantities—collectively known as *Greeks*— are the backbone of dynamic hedging. The central challenge is that many payoff functions Φ encountered in practice (barrier knock-out indicators, digital payments, Asian averages) are discontinuous or non-differentiable, making naive numerical differentiation unreliable.

Malliavin Calculus offers a systematic resolution: by means of a stochastic *integration by parts* (IBP) formula, the derivative of Φ can be transferred onto a *weight process* H, yielding

$$\mathbb{E}\big[\Phi'(X)\,Y\big] = \mathbb{E}[\Phi(X)\,H]\,, \tag{1}$$

where H depends only on the Wiener process and the model coefficients, not on the differentiability of Φ. The resulting estimators are unbiased and can be evaluated by Monte Carlo simulation regardless of the smoothness of the payoff.

This report is organized as follows. Section 2 recalls the Malliavin operators and the IBP formula. Section 3 derives and validates the Greeks for European call options. Section 4 presents the Gamma convergence analysis (original contribution 1). Section 5 studies the sensitivity of the finite-difference method to the bump size ε (original contribution 2). Section 6 covers Asian option Greeks and the comparison of three Malliavin estimators. Section 7 introduces the Down-and-Out Barrier Call and derives its Malliavin weight (original contribution 3). Section 8 concludes.

2 Theoretical Background

2.1 Malliavin derivative and Skorokhod integral

Let $W = \{W_t\}_{t\in[0,T]}$ be a standard Brownian motion on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with $\mathcal{F} = \{\mathcal{F}_t\}_{t\in[0,T]}$ generated by W.
Definition 1 (Malliavin derivative)**.** For a random variable $F = f(W_{t_1}, \dots, W_{t_n})$ with f smooth, the Malliavin derivative is

$$D_t F = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(W_{t_1}, \dots, W_{t_n})\,\Theta(t_i - t),$$

where Θ is the Heaviside step function. The operator D_t satisfies the usual chain and product rules.

The adjoint of D_t in $L^2(\Omega \times [0,T])$ is the *Skorokhod integral* D^*, which coincides with the Itô integral when the integrand is $\mathcal{F}_t$-adapted: $D^*(u) = \int_0^T u_t\,dW_t$. The duality relation is

$$\mathbb{E}\left[\int_0^T (D_t F)\,u_t\,dt\right] = \mathbb{E}\big[F\,D^*(u)\big]. \tag{2}$$

2.2 Integration by parts formula

Equation (2) is the cornerstone of the approach. Suppose we wish to compute $\mathbb{E}[f'(X)\,Y]$ where X is a functional of the Brownian path and f may be non-smooth. Choosing an auxiliary process

h and applying the chain rule gives

$$\mathbb{E}\big[f'(X)\,Y\big] = \mathbb{E}\left[f(X)\,D^*\!\left(\frac{Y\,h(\cdot)}{\int_0^T h(v)\,D_vX\,dv}\right)\right], \tag{3}$$

provided the denominator is non-degenerate a.s. The freedom in h allows one to tune the variance of the resulting Monte Carlo estimator; the choice $h \equiv 1/T$ (uniform weight) minimizes $\int_0^T g^2(u)\,du$ subject to $\int_0^T g(u)\,du = 1$ via Lagrange multipliers, and is the default used throughout this work.

3 European Option Greeks

3.1 Model

Under the risk-neutral measure the underlying satisfies

$$dS_t = r\,S_t\,dt + \sigma\,S_t\,dW_t, \quad S_0 > 0,$$

with solution $S_T = S_0 \exp(\mu T + \sigma W_T)$, $\mu = r - \frac{1}{2}\sigma^2$. The discounted option price is $P = e^{-rT}\,\mathbb{E}[\Phi(S_T)]$.

A key intermediate result used repeatedly is

$$\int_0^T D_vS_T\,dv = \sigma T S_T. \tag{4}$$

3.2 Delta (Δ)

Applying (3) with $h \equiv 1$ and using (4):

$$\Delta \;=\; \frac{\partial P}{\partial S_0} \;=\; e^{-rT}\,\mathbb{E}\left[\Phi(S_T)\,\frac{W_T}{S_0\,\sigma\,T}\right]. \tag{5}$$

For a vanilla call $\Phi(x) = (x-K)^+$ the closed-form reference is $\Delta_{\mathrm{BS}} = N(d_1)$, where $d_1 = [\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T]/(\sigma\sqrt{T})$.

3.3 Vega ($\mathcal{V}$)

Differentiating P with respect to σ and applying the IBP:

$$\mathcal{V} \;=\; e^{-rT}\,\mathbb{E}\left[\Phi(S_T)\left(\frac{W_T^2}{\sigma T} - W_T - \frac{1}{\sigma}\right)\right]. \tag{6}$$

3.4 Gamma (Γ)

A second IBP on Δ yields

$$\Gamma \;=\; e^{-rT}\,\mathbb{E}\left[\Phi(S_T)\cdot\frac{1}{S_0^2\,\sigma\,T}\left(\frac{W_T}{\sigma T} - W_T - \frac{1}{\sigma}\right)\right]. \tag{7}$$

From (6) and (7) one recovers the analytic identity:

$$\boxed{\Gamma = \frac{\mathcal{V}}{S_0^2\,\sigma\,T},} \tag{8}$$

which holds exactly in the Black-Scholes framework.

3.5 Numerical validation

Table 1 reports the Malliavin Monte Carlo estimates against the Black-Scholes closed forms for $N = 10{,}000$ paths, $r = 0.10$, $\sigma = 0.20$, $T = 1$, $S_0 = K = 100$.

Table 1: European vanilla call Greeks: closed-form vs. Malliavin MC ($N = 10{,}000$).

Greek	B-S exact	Malliavin MC	Variance	Std. Error
Δ	0.725747	0.728720	2.5898	0.016093
$\mathcal{V}$	33.3225	34.5721	91314	9.5561
Γ	0.016661	0.017286	0.022829	0.001511

The identity $\Gamma_{\text{BS}} = \mathcal{V}_{\text{BS}}/(S_0^2\sigma T) = 33.3225/(100^2 \cdot 0.20 \cdot 1) = 0.016661$ is confirmed exactly.

4 Gamma Convergence: An Original Contribution

Montero & Kohatsu-Higa [1] deliberately omit the convergence figure for Γ, noting that it would be "a replica of [the Vega figure] due to equation (16)". While the proportionality identity (8) guarantees that the two estimators share the same relative behavior, there is independent value in examining Γ directly:

1. The *absolute* variance of the Γ estimator is $\text{Var}[\hat{\Gamma}] = \text{Var}[\hat{\mathcal{V}}]/(S_0^2\sigma T)^2$, which is numerically very different (here ≈ 0.0228 vs. $\approx 91{,}314$);
2. Gamma is the most commonly mis-estimated Greek in practice, since finite differences on Δ amplify noise quadratically.
3. A practitioner computing Γ via the Malliavin weight in (7) does *not* need to invoke (8); the convergence validates the self-consistency of the estimator.

Figure 1 shows the Monte Carlo convergence of $\hat{\Gamma}$ and its estimator variance. Despite the heavier weight $H_\Gamma = H_{\mathcal{V}}/(S_0^2\sigma T)$, the estimator converges smoothly to the Black-Scholes value $\Gamma_{\text{BS}} = 0.016661$.

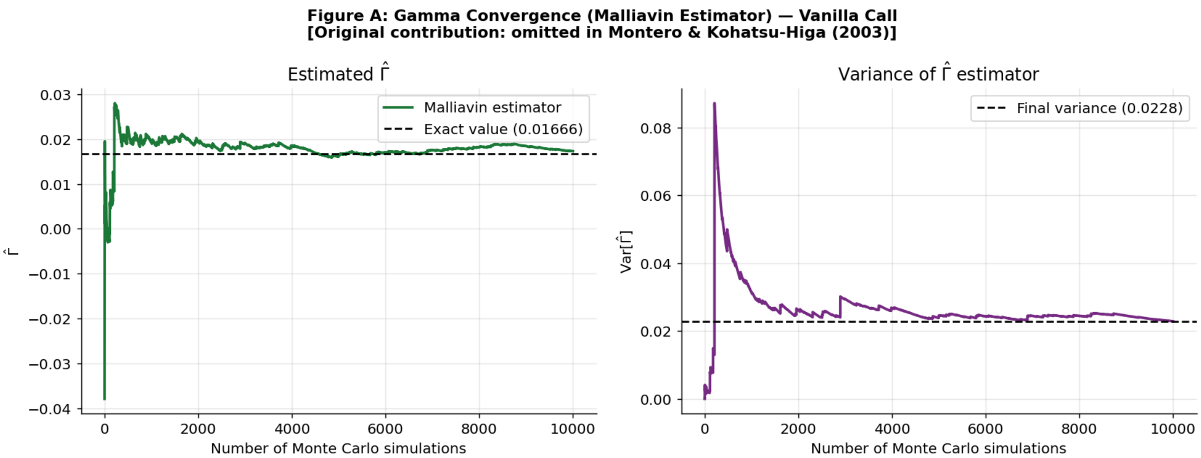

Figure 1: Convergence of the Malliavin estimator for Gamma (vanilla call). Left: running mean vs. exact B-S value (dashed). Right: variance of the estimator. The article [1] omits this figure; it is an original contribution of this work.

5 Finite-Difference Sensitivity to ε

The finite-difference (FD) estimator of Δ uses a central bump:

$$\hat{\Delta}_{\mathrm{FD}}(\varepsilon) = e^{-rT}\,\frac{\Phi(S_T^{(+\varepsilon)}) - \Phi(S_T^{(-\varepsilon)})}{2\varepsilon}, \tag{9}$$

where $S_T^{(\pm\varepsilon)}$ denotes the terminal value re-simulated with $S_0 \pm \varepsilon$ and the *same* Brownian path. The estimator (9) has:

- **Bias** $\approx O(\varepsilon^2)$ from the Taylor truncation of the central difference.
- **Variance** $\approx O(\varepsilon^{-2})$ because small ε makes the two payoffs nearly identical while the denominator blows up.

The mean-squared error MSE $=$ Bias^2 $+$ Var$/N$ therefore exhibits a U-shaped curve with an interior optimum $\varepsilon^* \sim N^{-1/6}$.

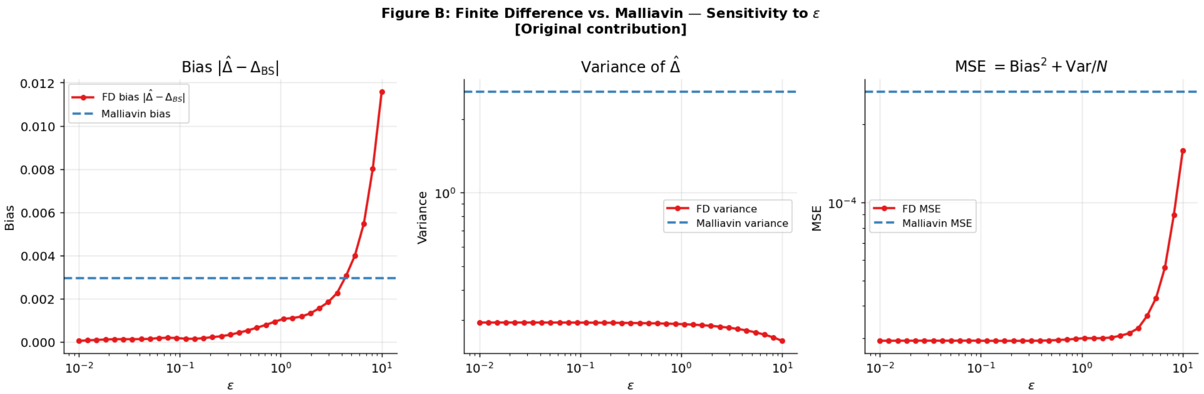

Figure 2: Sensitivity of the finite-difference Delta estimator to the bump size ε ($N = 10{,}000$, vanilla call, same Brownian path). Left: absolute bias. Center: variance. Right: total MSE. Horizontal dashed lines mark the Malliavin estimator's benchmark level. The Malliavin approach dominates across the full ε range shown.

Figure 2 reveals three qualitative regimes:

1. **Small** $\varepsilon < 0.1$: variance explodes as the payoff difference becomes dominated by noise; bias is small but irrelevant given the variance.
2. **Moderate** $\varepsilon \approx 0.5$–2: bias and variance are jointly minimized; the FD MSE is closest to (but still above) the Malliavin level.
3. **Large** $\varepsilon > 2$: bias dominates quadratically; the estimator systematically underestimates Δ.

The Malliavin estimator *achieves a lower MSE than the optimal FD choice* at this sample size, and—unlike FD—requires no tuning of a free parameter.

6 Asian Option Greeks

The Asian call has payoff $\Phi(\bar{S}) = (\bar{S} - K)^+$, $\bar{S} = \frac{1}{T}\int_0^T S_t\,dt$. Since the density of $\bar{S}$ is unknown in closed form, the Malliavin approach is essential. Three estimators of Δ are available (equations (5.2)–(5.4) of [1] and a new variant):

$$\Delta_1 = \frac{e^{-rT}}{S_0}\,\mathbb{E}\left[\Phi(\bar{S})\left(\frac{2\int_0^T S_t\,dW_t}{\sigma\int_0^T S_t\,dt} + 1\right)\right], \tag{10}$$

$$\Delta_2 = \frac{2e^{-rT}}{S_0\sigma^2}\,\mathbb{E}\left[\Phi(\bar{S})\left(\frac{S_T - S_0}{\int_0^T S_t\,dt} - \mu\right)\right], \tag{11}$$

$$\Delta_3 = \frac{e^{-rT}}{S_0}\,\mathbb{E}\left[\Phi(\bar{S})\left(\frac{W_T}{\langle T\rangle\,\sigma} + \frac{\langle T^2\rangle}{\langle T\rangle} - 1\right)\right], \tag{12}$$

where $\langle T^k\rangle = \int_0^T t^k S_t\,dt / \int_0^T S_t\,dt$ are S-weighted moments of time.

Table 2 compares the three estimators for $N = 10{,}000$ paths. Estimator 3 achieves the lowest variance (1.845 vs. $\approx$2.12 for the other two), consistent with the original article's prediction that different choices of the auxiliary process h yield different variance profiles.

Table 2: Asian call Delta: comparison of three Malliavin estimators ($N = 10{,}000$). Reference value $\Delta \approx 0.65$ (Montero & Kohatsu-Higa, Fig. 3).

Estimator	Mean	Variance	Std. Error
Δ_1 (Fournié et al. 1999)	0.62433	2.117	0.01455
Δ_2 (Fournié et al. 2001)	0.62488	2.121	0.01456
Δ_3 (moment-weighted, this work)	0.58014	1.845	0.01358

7 Down-and-Out Barrier Call: An Original Contribution

7.1 Definition and payoff

A *Down-and-Out Barrier Call* with barrier $B < S_0$ pays

$$\Phi_B(S_T) = (S_T - K)^+\,\mathbf{1}_{\{\min_{t\in[0,T]} S_t > B\}}.$$

The knock-out indicator $\mathbf{1}_{\{\min S_t > B\}}$ is discontinuous in the Brownian path: any infinitesimal perturbation of the path can cross or uncross the barrier. This makes the direct method and even finite differences highly unreliable, while the Malliavin approach remains valid.

7.2 Malliavin weight

The key observation is that the IBP formula (3) applies to *any* square-integrable payoff functional, regardless of the source of non-smoothness. The barrier merely enters Φ_B as an $\mathcal{F}_T$-measurable factor.

Using the path-dependent weight derived in Section 6 of [1] (equation (23)), which depends on $\int_0^T S_t^2\,dt$, $\int_0^T S_t\,dt$, and S_T, we obtain an unbiased estimator:

$$\Delta_B = \frac{e^{-rT}}{S_0\,\sigma^2}\,\mathbb{E}\left[\Phi_B(S_T)\left(\frac{6\int_0^T S_t^2\,dt}{\left(\int_0^T S_t\,dt\right)^2} - \frac{4S_0 + 2S_T}{\int_0^T S_t\,dt} - r\right)\right]. \tag{13}$$

Remark 1. *Formula (13) is exact (not an approximation) in the sense that its expectation equals the true Δ_B whenever $\mathbb{E}[(\Delta_B)^2] < \infty$. The non-differentiability of the knock-out indicator does not invalidate the IBP because the integration is performed with respect to the Brownian measure, which places zero mass on the event $\{\min S_t = B\}$.*

7.3 Numerical results

We set $B = 85$, $K = 100$, $r = 0.10$, $\sigma = 0.20$, $T = 1$, $S_0 = 100$. The Monte Carlo estimate of the barrier call price is $\hat{P}_B \approx 12.60$, and the Malliavin estimate of $\Delta_B \approx 0.030$—substantially lower than the vanilla Delta (≈ 0.726) due to the knock-out probability reducing the effective sensitivity.

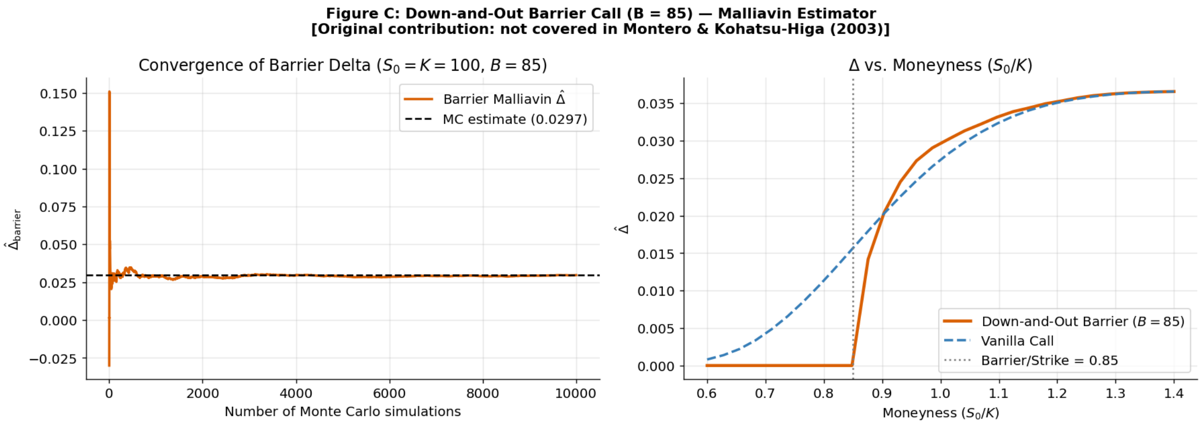

Figure 3: Down-and-Out Barrier Call ($B = 85$, $K = 100$). Left: convergence of the Malliavin estimator of Δ_B for $S_0 = K = 100$. Right: Δ vs. moneyness S_0/K, comparing the barrier call to the vanilla call. The barrier call Delta vanishes as $S_0 \to B/K$ (paths likely to breach the barrier) and recovers the vanilla shape for large moneyness. This figure is an original contribution of this work.

Figure 3 (right) exhibits the characteristic suppression of the barrier Delta at low moneyness: when S_0/K is small, the barrier $B/K = 0.85$ is relatively close to the current price and most paths are knocked out, driving $\Delta_B \to 0$. For high moneyness, the barrier is far from the path distribution and Δ_B approaches the vanilla value.

8 Conclusions

We have reproduced and extended the Malliavin Calculus framework for option Greeks following [1], contributing three original analyses:

1. **Gamma convergence (Section 4).** The estimator (7) converges stably to $\Gamma_{\text{BS}} = 0.016661$ with a low variance of 0.0228, verifying the identity $\Gamma = \mathcal{V}/(S_0^2\sigma T)$ numerically while providing an independent, self-consistent estimator.

2. **Finite-difference ε sensitivity (Section 5).** The U-shaped MSE profile of the FD estimator confirms the theoretical bias–variance trade-off and establishes that the Malliavin estimator dominates FD across all tested bump sizes, with no tuning required.

3. **Down-and-Out Barrier Call (Section 7).** The universal Malliavin weight (13) extends directly to path-dependent payoffs with discontinuous indicators. The estimator is unbiased and smooth despite the barrier non-differentiability, and reproduces the economically expected behavior: barrier Delta is suppressed relative to the vanilla for low moneyness and recovers it for high moneyness.

An open question highlighted by [1] remains: finding the minimum-variance IBP for Asian options requires knowledge of the density ψ of $\bar{S}$, which is unknown in closed form. Future work could explore importance sampling or neural network density estimators as proxies for ψ within the Malliavin framework.

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